

Student Performance Analysis

Cognevance Technologies — Level 1 Data Science & Data Analytics Project

1. Introduction

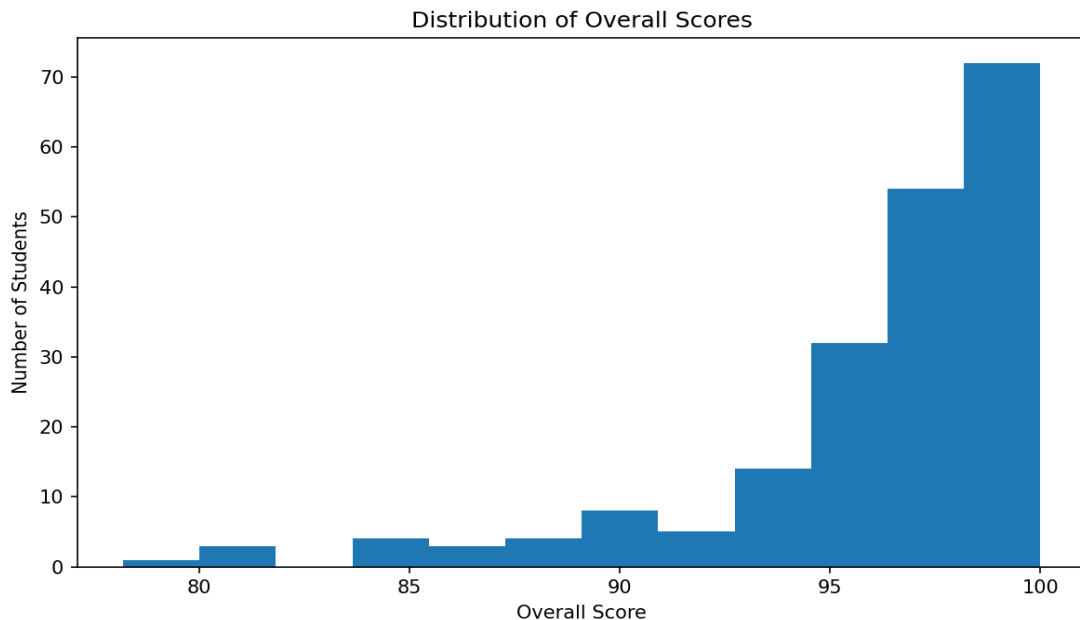
This project analyzes a synthetic dataset of 200 students to identify patterns in academic performance, attendance, study habits, assignments, and grades.

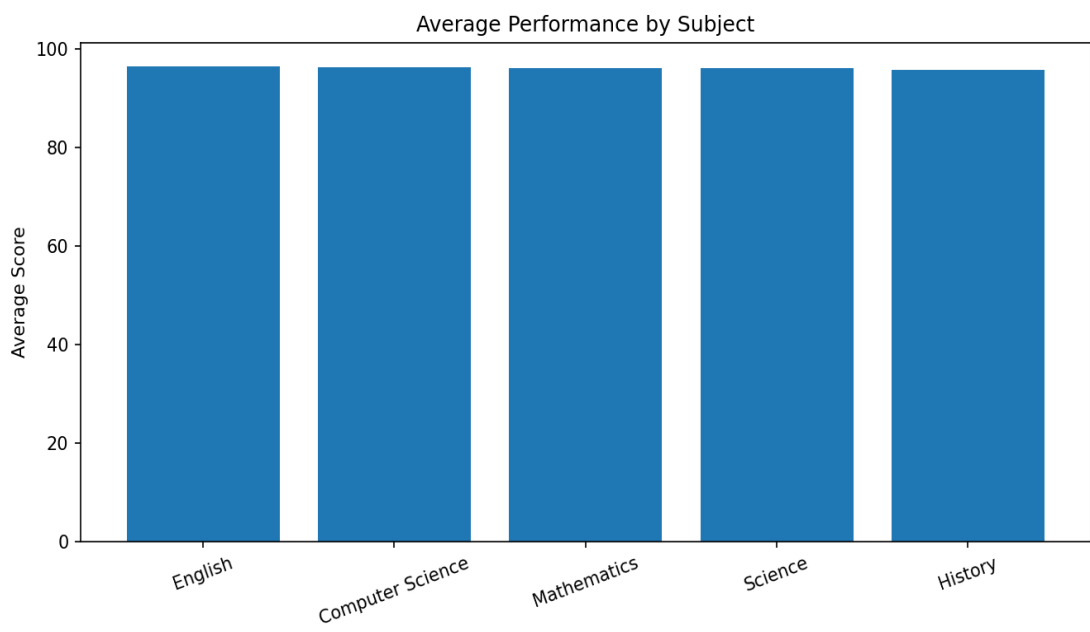
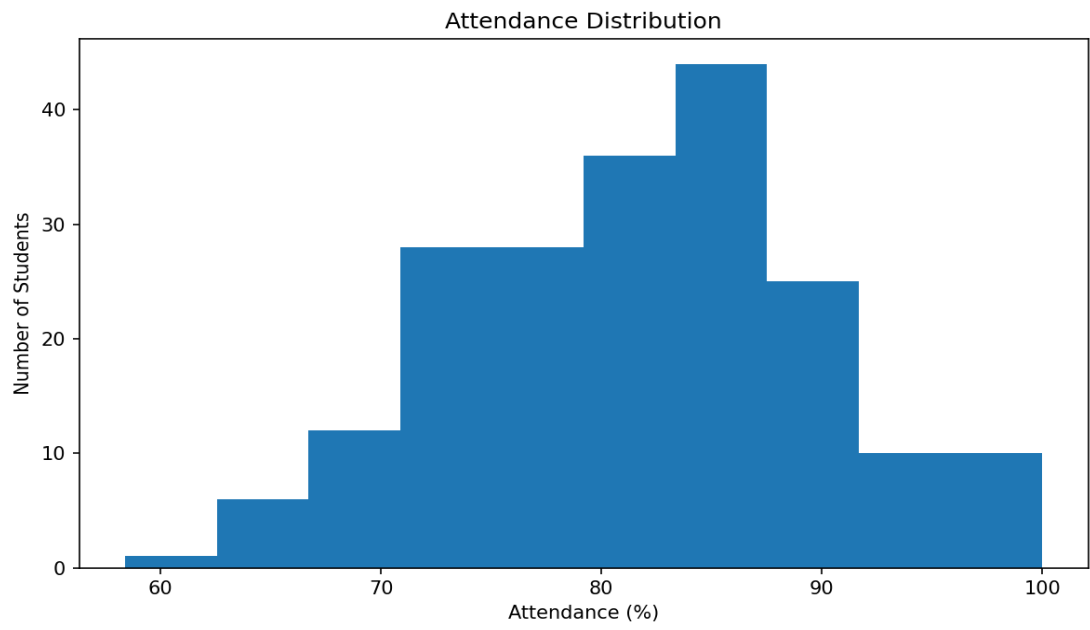
2. Methodology

The dataset was loaded with pandas, checked for duplicates and missing values, summarized with descriptive statistics, and analyzed using subject averages and correlation. Matplotlib was used to create charts.

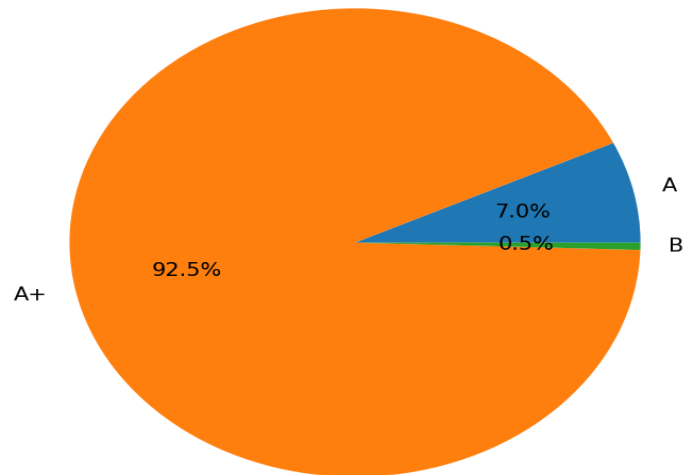
3. Key Findings

The attendance-to-overall-score Pearson correlation is 0.260, showing a positive association in this synthetic dataset. The subject with the highest average score is English (96.4). The overall average score is 96.1.

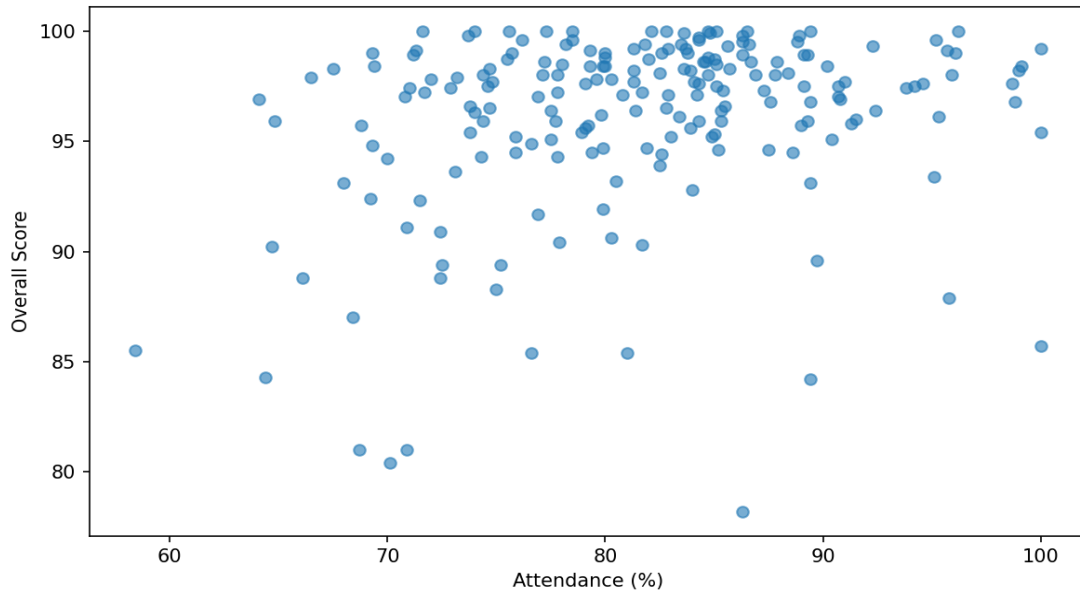




Grade Distribution



Attendance vs Overall Score



4. Recommendations

Schools can monitor attendance alongside academic scores, provide targeted support for students with lower grades, encourage consistent study and assignment completion, and use subject-level performance to identify areas requiring additional teaching support.

5. Conclusion

The analysis demonstrates a complete beginner-friendly data analytics workflow: data preparation, exploratory analysis, visualization, relationship analysis, and actionable recommendations.